Probability as a Model

November 11, 2026

From a probability table to a model

In Unit 7, we started with a probability table. We knew the five possible service-call outcomes and their probabilities, and used them to answer questions.

What if we only have the data?

- We have records for 600 moving jobs. We need to estimate the probabilities.
- We can already describe these jobs with a mean, a standard deviation, or a histogram.
- Now we will choose a model for how the outcomes vary, fit it to the data, and check whether it works well enough for our question.

Key idea: what a probability model is

A probability model describes possible outcomes and how likely they are. We choose one based on how we think the data arise, then estimate its parameters and check its fit.

Working with a probability model

1. **Choose a model.** Think about what we are measuring and how the outcomes arise.
2. **Fit it.** Find parameter values that agree well with the data.
3. **Check it.** Compare the model's predictions with what we observed. Revise the model if the differences matter.
4. **Use it.** Estimate a probability or make a plan for a future outcome.

For example, past moving jobs can help us estimate the chance that a new job will run into overtime. That estimate gives a dispatcher something useful to plan around.

What this unit gives you

By the end, you can

- choose a model for a count, a waiting time, or a total
- estimate its parameters and explain the estimates
- check a fit and recognize when it needs work
- use a distribution to calculate a probability
- connect a prediction interval to a forecast distribution

Common mistakes to watch for

- choosing a distribution from a histogram alone
- using a fitted model without checking it
- overlooking extra variation in count data
- treating $\exp(\hat{y})$ from a logged model as the mean
- giving a time estimate without any uncertainty

Connections to earlier units

- **Unit 1** introduced the 68–95–99.7 rule. We will see where those percentages come from in the normal distribution.
- **Unit 4** gave a prediction interval of 10.74 to 14.04 hours for one moving job. We will look at the distribution behind that interval and explain its width.
- **Units 4 and 5** used AIC to compare models. We will introduce *likelihood*, the measure of fit that appears in its formula.

In practice: filling in the details

You have already used these tools. Now we can explain how they work and when their assumptions are reasonable.

Choosing a distribution

Before choosing a distribution, ask:

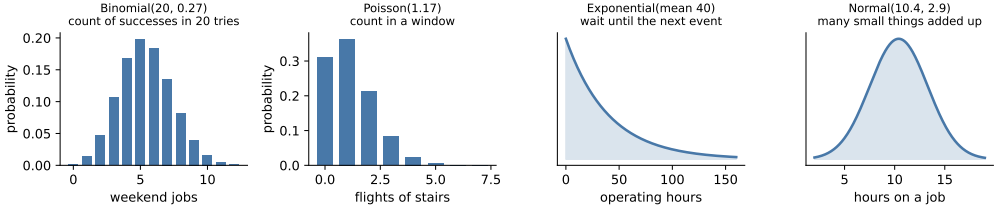
1. **What are we measuring?** Successes out of a fixed number of tries, events in a time period, a waiting time, or a total made up of many smaller amounts?
2. **What assumptions does the model need?** Are those assumptions reasonable here?

Then use the data to see how well the proposed model fits.

Watch for: relying on the histogram alone

Different processes can produce similar histograms. With only 40 observations, the shape can also be hard to judge. Consider how the data arise as well as how they look.

Four distributions we will use



Each distribution describes a different kind of outcome. We will look at its assumptions, estimate its parameters, and use it to answer a practical question.

Binomial: successes in a fixed number of tries

The **binomial**(n, p) counts successes in a fixed number of independent tries. Each try has the same probability of success.

- **Parameters.** n is the number of tries. We estimate the success probability p.
- **Mean and spread.** $\mathbb{E}[X] = np$ and $\mathrm{SD}(X) = \sqrt{np(1-p)}$. The SD formula follows from adding variances, as we did in Unit 7.
- **Moving jobs.** Of the 600 jobs, **163 ran on a weekend**. A binomial model for the weekend count assumes independent bookings with the same weekend probability.

Watch for: checking the booking assumptions

A promotion or holiday may change the weekend probability. Several jobs booked by the same customer may be related. Either can make the binomial model less suitable.

Planning for weekend jobs

Twenty jobs are booked for next week. How much weekend capacity might we need?

Question	**Answer**
How many weekend jobs on average?	$20 \times 0.272 = 5.43$
How much does that vary?	$\sqrt{20(0.272)(0.728)} = 1.99$
$\mathbb{P}$(exactly 5)	0.198
$\mathbb{P}$(8 or more)	**0.150**

- The average is 5.43 weekend jobs, but the count varies. Eight or more occur about one week in seven under this model.
- We can plan for a new week with 20 bookings using the weekend rate estimated from earlier jobs.

Poisson: events in a time period or area

The **Poisson**(λ) models a count over a defined period, area, or other unit. It assumes independent events at a constant rate, with no fixed upper limit on the count.

- **Parameter.** λ is the expected count per unit.
- **Mean and variance.** $\mathbb{E}[X] = \lambda$ and $\mathrm{Var}(X) = \lambda$. Their equality gives us a useful way to check the model.
- **Moving jobs.** Flights of stairs per job have mean **1.168** and variance **1.182**. Those are close, so a Poisson model is worth checking further.

In practice: where this one shows up

Possible uses include defects per batch, arrivals per hour, insurance claims, and typos per page. We still need to check the assumptions in each setting.

Using the Poisson model for stairs

With the fitted mean $\hat{\lambda} = 1.168$, we can calculate:

	Model probability	**Observed proportion**
$\mathbb{P}$(no stairs at all)	0.311	0.312
$\mathbb{P}$(three or more flights)	0.114	0.113

- We fitted the model using the mean. It also closely matches these two observed proportions, which gives us more reason to use it.
- The fitted distribution lets us estimate probabilities for future jobs, including counts that were rare in our sample.

Key idea: checking more than the fitted mean

A useful check asks whether the model also captures features we did not directly fit.

Exponential: waiting for the next event

The **exponential** models a waiting time, with rate λ and mean wait $1/\lambda$.

- **Connection to the Poisson.** For a Poisson process with rate λ, the waiting times between events are exponential with mean $1/\lambda$.
- **Example.** Suppose a truck breaks down once per 40 operating hours on average. Under this model, $\mathbb{P}(\text{no breakdown in the next 10 hours}) = e^{-10/40} = 0.78$, and the chance it runs for 40 hours without a breakdown is $e^{-1} = 0.37$.
- **Memoryless.** After 30 hours without a breakdown, the model gives the same probability of surviving the next 10 hours as it did at the start.

Watch for: memorylessness is a strong assumption

If parts wear out, the chance of a breakdown can increase with age. An exponential model assumes a constant failure rate, so it may be a poor choice for that situation.

Normal: many small contributions added together

The **normal**(μ, σ^2) often approximates a sum of many small, roughly independent contributions.

- A moving job includes time for loading, driving, and unloading, along with smaller tasks and delays. A normal approximation may work if no single part dominates.
- Unit 7 used simulation to show how averages become approximately normal. The same reasoning can apply to sums, including total time.
- **In the data.** Hours have mean **10.43**, standard deviation **2.94**, and a mild right skew of 0.36.

Key idea: when a normal approximation is reasonable

Look for many contributions that add together, with no single contribution dominating. Strong dependence or a few very large contributions can make the approximation poor.

Which distribution would you consider?

Quick check: suggest a model and its parameters

1. Of 300 quotes sent this month, how many turn into bookings?
2. How many support tickets arrive between 9 and 10 on a Tuesday?
3. How long until the next support ticket arrives?
4. What is the total weight of 200 boxes on a truck?

Which distribution would you consider?

Quick check: suggest a model and its parameters

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4. What is the total weight of 200 boxes on a truck?

1. Binomial, with $n = 300$ and an unknown booking probability p.
2. Poisson, with λ equal to the expected number of tickets per hour.
3. Exponential, with the same arrival rate λ and mean wait $1/\lambda$.
4. Normal, with a mean and variance for the total, if no box dominates the weight.

Could more than one model make sense?

Consider two possible models for flights of stairs:

- **Poisson.** Treat flights per job as a count with a common rate and no fixed upper limit.
- **Binomial.** Suppose each job has a fixed number of opportunities to encounter stairs, each with the same probability. That would put an upper limit on the count.

The assumptions differ, and so do the predictions. The binomial variance is *below* its mean. The Poisson variance *equals* its mean. Here we observed 1.182 and 1.168, respectively, which is consistent with the Poisson relationship.

Key idea: using context and data together

How the jobs work helps us choose plausible models. The data help us check those choices.

The assumptions behind Poisson(1.168)

Using the same Poisson model for every job assumes:

- jobs have the same expected count
- events occur independently
- the variance equals the mean
- there is no fixed upper limit, so even 9 flights has some probability

Different building types or a crew that stops counting at 6 could make these assumptions unreasonable. A poor fit may mean we need different assumptions, rather than a better estimate of the same parameter.

Your turn

Your turn: three minutes, in pairs

A warehouse records the number of orders received each day, how many shipped late, and the minutes between the first two orders of the morning. It has records for 90 days.

For each measurement, suggest a distribution, identify its parameters, and give one assumption you would need to check. Which model are you least confident about, and why?

Fitting a model with likelihood

Once we have chosen a distribution, we still need its parameters: p, λ, or μ and σ.

- **Try a value.** Suppose $p = 0.20$. For the weekend data, this gives a probability to the outcomes we observed.
- **Calculate a score.** We call that probability the **likelihood**, $\mathcal{L}(p)$. It tells us how well this value of p accounts for the data.
- **Find the highest score.** The value with the largest likelihood is the **maximum likelihood estimate**, $\hat{p}$.

Key idea: the idea behind maximum likelihood

Choose the parameter value that gives the observed data the highest probability under the model.

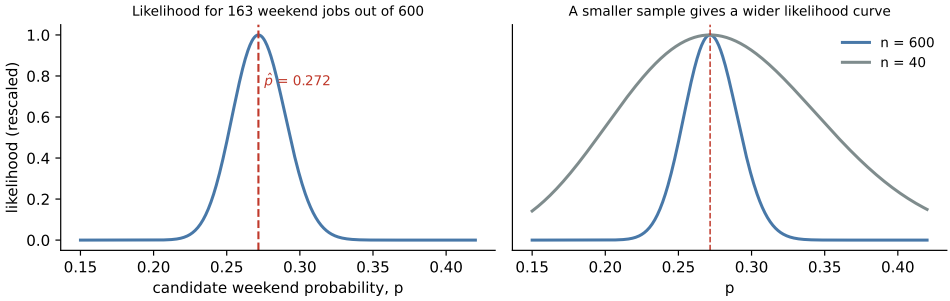
Likelihood for the full dataset

For independent jobs, we multiply the probabilities of their observed weekend or weekday outcomes:

$$\mathcal{L}(p) \;=\; \prod_{i=1}^{600} \mathbb{P}(\text{job } i \text{ came out as it did}) \;=\; p^{163}(1-p)^{437}.$$

- There are 163 weekend jobs and 437 weekday jobs. Each contributes a factor of p or $1-p$ to the product.
- This multiplication uses our independence assumption. Related bookings would need a model that accounts for their dependence.
- Products of many small probabilities can be difficult to compute accurately. Software usually maximizes $\log \mathcal{L}$, which turns the product into a sum and keeps the same peak.
- This is the *log-likelihood* you see in model output and in AIC.

Likelihood for the weekend data



Left: the likelihood for 163 weekend jobs out of 600. It compares candidate values of p. Right: an illustration of how much wider the curve would be with only 40 jobs at the same weekend proportion. These curves are likelihoods, rather than probability distributions for p.

Familiar estimates from likelihood

For the binomial, maximizing $\mathcal{L}(p) = p^{163}(1-p)^{437}$ gives

$$\hat{p} = \frac{163}{600} = 0.2717,$$

the sample proportion. For the Poisson, maximizing the likelihood gives $\hat{\lambda} = \bar{x}$, the sample mean.

- In these cases, maximum likelihood gives estimates that already seem sensible.
- The same method also works for more complicated models, where the best parameter values are harder to find by inspection.

In practice: where you will see it again

Logistic regression, Poisson regression, and many survival models use likelihood to estimate their parameters. Software does the numerical search for us.

Peak width and standard errors

The two curves have the same $\hat{p}$, but different widths.

- **A narrow peak:** nearby parameter values fit much less well. The estimate is relatively precise.
- **A broad peak:** nearby values fit almost as well. There is more uncertainty about the parameter.
- The curvature of the log-likelihood near its peak gives an approximate standard error. Here, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n} = \mathbf{0.0182}$, and the 95% interval runs from **0.236 to 0.307**.

Key idea: another way to estimate uncertainty

Unit 1 used formulas and Unit 6 used resampling. Likelihood curvature gives us another route to a standard error, including for models without a simple formula.

Estimating the Poisson rate

For stairs, $\hat{\lambda} = \bar{x} = 1.168$, with $\widehat{\mathrm{SE}}(\hat{\lambda}) = \sqrt{\hat{\lambda}/n} = \sqrt{1.168/600} = 0.044$.

- We report the estimated rate along with its standard error.
- The observed counts for 0, 1, 2, 3, 4, 5 and 6 flights take *seven* numbers to list. A Poisson model uses one parameter to describe the whole distribution.
- That simpler description can give precise estimates when the model fits. If its assumptions are wrong, the estimates can be misleading. We need to check the fit.

The likelihood in AIC

Units 4 and 5 compared models using

$$\text{AIC} = -2 \log \mathcal{L} + 2k.$$

- $\log \mathcal{L}$ measures fit at the model's estimated parameter values.
- k is the number of parameters. The $2k$ term penalizes complexity, since adding parameters can improve fit even without improving predictions.
- We prefer a lower AIC. Better fit lowers the first term, while extra parameters increase the second.

In practice: comparing the same outcome

Compare AIC for the same observations on the same outcome scale. Models of y and $\log y$ need a likelihood adjustment before comparing their reported AIC values.

Simple estimates in software

Task	How
Binomial p	`x.sum() / n`
Poisson λ	`x.mean()`
Normal mean and sample SD	`x.mean()`, `x.std(ddof=1)`
Exponential rate	`1 / x.mean()`
Other distributions	`scipy.stats.<dist>.fit(x)`
The log-likelihood of a fit	`fit.llf` in statsmodels

Watch for: an estimate still needs a fit check

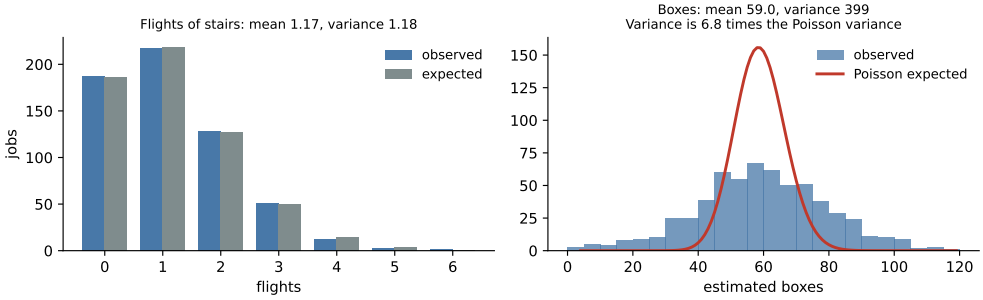
A fitting function can return parameter estimates even when the distribution is a poor match for the data. Getting an answer from the software is only the fitting step.

Checking the fit

- **For counts,** compare the observed counts with the counts the fitted model expects. A table or plot can make differences easy to see.
- **For continuous outcomes,** use a Q-Q plot to compare the observed shape with the proposed distribution, including its tails.

In both cases, we ask: *are the data reasonably consistent with this model, especially in the parts we need for our question?*

A close fit and a poor fit



Left: flights of stairs compared with Poisson(1.168). Right: estimated boxes compared with Poisson(59.0). We used the same fitting method, but the model fits the stairs much better than the boxes.

Observed and expected stair counts

Flights	0	1	2	3	4	5	6
Observed	187	217	128	51	12	3	2
Poisson expects	186.5	217.9	127.3	49.6	14.5	3.4	0.8

- The fitted rate reproduces the overall pattern of counts quite well.
- At six flights, we observed two jobs where the model expected 0.8. That is a large ratio, but a small difference in jobs. It gives little reason on its own to reject the model.

Watch for: interpreting a goodness-of-fit test

Large samples can make small departures statistically significant. Along with any p-value, consider the size of the mismatch and whether it would change your answer.

When the variance is too large for a Poisson

Estimated boxes per job have mean **59.0** and variance **399**.

- A Poisson model has equal mean and variance. Here, the variance is **6.8 times** the mean, so the model misses a substantial amount of variation.
- We call this **overdispersion**: more variation than the count model allows.
- One possible explanation is that jobs differ in size. Combining large and small jobs produces more variation than a model with one common rate can capture.

Key idea: a useful first check for Poisson counts

Compare the sample variance with the sample mean. A ratio far from one suggests that a Poisson model with one common rate may be unsuitable.

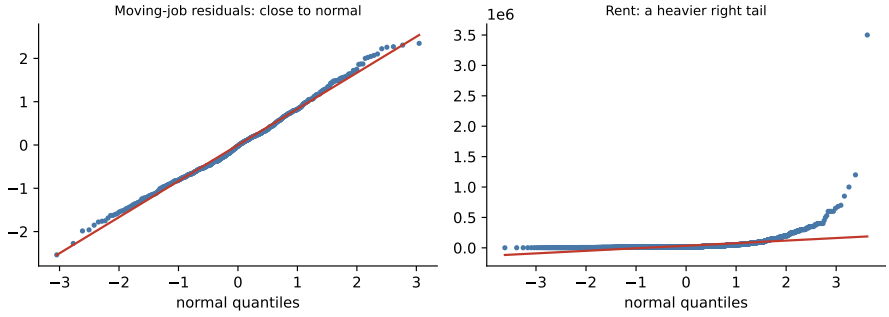
Why overdispersion matters for regression

- Extra variation is common in count data. We will return to it when we fit count regressions in Unit 9.
- A Poisson regression can give standard errors that are too small when the data have more variation than the model allows. Effects may then look more certain than they are.
- Options include quasi-Poisson, which adjusts the standard errors, and negative binomial regression, which includes a parameter for extra variation.

In practice: what to report now

Start by reporting what you found: the variance is 6.8 times the mean. That observation explains why a simple Poisson model needs more work.

Reading a normal Q-Q plot



Sort the observations and compare them with corresponding normal quantiles. Points close to a straight line suggest a similar shape. Systematic bends show where the shapes differ. Here we compare moving-job residuals and rent with a normal distribution.

What the bends can tell us

What you see	Possible explanation
Straight in the middle, both ends curling up	Right skew, with a long upper tail
Both ends bending away from the line	Heavier tails than normal
Both ends bending toward the line	Lighter tails than normal, possibly a floor or ceiling
A flat stretch of points	Repeated values, rounding, or a cap

- Rent bends upward because a few properties cost much more than a typical one. In Unit 1, we saw a mean of \$35,010 and a median of \$16,000. The plot shows that same right skew.

Does the mismatch matter for our question?

A model can still be useful when it is an imperfect match. We need to consider what we plan to calculate.

- **Estimating a mean:** mild skew in 600 jobs may cause little difficulty. The averaging argument from Unit 7 helps explain why.
- **Estimating a high percentile:** a heavy right tail can matter a great deal. A normal model may underestimate the capacity we need.
- **Predicting one job:** a large sample does not make the individual outcome normal. The distribution's shape still matters for its prediction interval.

Key idea: a question to ask of any fit

Which part of the distribution does my answer depend on, and does the model fit well there?

A routine for checking models

- For Poisson counts, compare variance with mean, then observed with expected counts.
- For continuous data, read the Q-Q plot, paying attention to the tail your question uses.
- Check more than the statistic used to fit the model. A fitted mean matching the sample mean gives little extra information.
- Report the evidence, even when the fit looks good. For example: "The variance was 1.18 and the mean was 1.17, consistent with a Poisson model."

Watch for: using a model beyond the checks you made

A model may fit the middle of the data well and still miss the tails. Check the part you need before reporting a tail probability.

Density and probability

For a continuous outcome, $\mathbb{P}(X = 10.43) = 0$. A single exact value has zero probability, even though an interval around it can have positive probability.

- The height of the curve is the **density** at that value.
- A probability is an **area under the curve** over an interval.
- Useful questions include the chance of taking less than 12 hours, more than 12 hours, or between two times. If we record hours rounded to a fixed precision, each recorded value represents a small interval.

Watch for: a density can be bigger than 1

The height can exceed 1. Probability comes from area, and the total area under the curve is 1.

The normal distribution

$$X \sim \text{Normal}(\mu, \sigma^2)$$

- μ is the mean and σ is the standard deviation. The curve is symmetric, and these parameters determine its location and spread.
- The **68–95–99.7 rule from Unit 1** describes areas within one, two, and three standard deviations of the mean.
- We can now see why that rule depends on shape. Those percentages come from the normal distribution, so they need not hold for every dataset.

In practice: checking the moving jobs

We observed **68.7%** of jobs within one SD of the mean and **95.7%** within two. Those percentages are close to what a normal model predicts.

Using a z-score

$$z = \frac{x - \mu}{\sigma}$$

- A z-score tells us how many standard deviations a value lies above or below the mean.
- For a 16-hour job, $z = (16 - 10.43)/2.94 = 1.89$. That is nearly two standard deviations above the mean.
- If the outcome is normal, standardizing lets us use the standard normal distribution to calculate probabilities.

Watch for: a z-score alone does not give a probability

A z of 3 can have a different tail probability in a skewed distribution than in a normal one. Check the shape before using normal probabilities.

Probabilities and percentiles in software

Question	Code
$\mathbb{P}(X < a)$	`norm.cdf(a, mu, sigma)`
$\mathbb{P}(X > a)$	`norm.sf(a, mu, sigma)`
$\mathbb{P}(a < X < b)$	`norm.cdf(b,...) - norm.cdf(a,...)`
The value with 90% below it	`norm.ppf(0.90, mu, sigma)`
$\mathbb{P}(Y > a)$ for a lognormal	`lognorm.sf(a, s=sigma, scale=np.exp(mu))`

In practice: matching the code to the question

Use `cdf` for the area below a value, `sf` for the area above it, and `ppf` to find a value from a given probability. For the lognormal, μ and σ describe $\log Y$.

How often does a job run past 16 hours?

Suppose we model hours as Normal(10.43, 2.94^2):

z for 16 hours	$(16 - 10.43)/2.94 = 1.89$
$\mathbb{P}(X > 16)$ from the model	**2.9%**
Observed proportion of the 600 jobs	**3.5%**

- The estimates are close, but the model is a little low. This is consistent with the mild right skew: the observed upper tail is somewhat heavier than the normal model's.
- Whether the difference matters depends on the decision. It may have little effect on a rough staffing plan, but matter more when every job over 16 hours incurs a penalty.

A quick check on continuous probabilities

Quick check: true or false? Explain your answer

1. A density of 1.4 at $x = 3$ means a 140% chance of landing near 3.
2. $\mathbb{P}(X \geq 16)$ and $\mathbb{P}(X > 16)$ differ for a continuous outcome.
3. A z of 2.5 always means the value is in the top 0.6% of the data.

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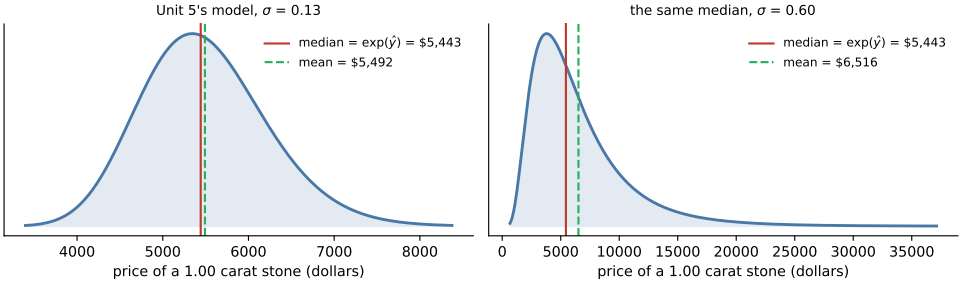
1. False. Density is height. To calculate a probability, we need the area over a specified interval.
2. False. The probabilities are equal because a single point contributes zero area.
3. False. That tail probability comes from the normal distribution. A z-score by itself does not tell us the shape of the distribution.

The lognormal distribution

If $\log(Y)$ is normal, then Y is **lognormal**. Its values are positive and right-skewed. Multiplicative effects can make a lognormal model worth considering.

- Prices, incomes, and durations are possible examples. Being positive alone does not tell us that a lognormal model fits.
- In Unit 5, we modeled the logged outcome. If the residuals are normal with constant spread on that scale, the conditional outcome distribution is lognormal.
- After predicting $\widehat{\log y}$, we still need to interpret $\exp(\widehat{\log y})$. Does it give the mean or the median?

The mean and median after exponentiating



For a lognormal model, exponentiating the predicted log outcome gives the **median**. Exponentiation preserves order, so it preserves the middle value. The **mean** is larger by a factor of $\exp(\sigma^2/2)$, where σ is the residual SD on the log scale.

How much does the mean differ from the median?

Residual SD σ (log scale)		Mean / median	
0.13	(Unit 5's diamond model)	1.009	$+0.9\%$
0.20		1.020	$+2.0\%$
0.40		1.083	$+8.3\%$
0.60		1.197	$+19.7\%$
0.80		1.377	$+37.7\%$

- The gap is small when the residual spread is small. It grows as the spread increases.
- Use the mean for a total budget and the median to describe a typical job.

Watch for: labeling the prediction correctly

To report the mean, multiply $\exp(\hat{y})$ by $\exp(\sigma^2/2)$.

Your turn

Your turn: two minutes

A logged model of repair cost has normally distributed residuals with standard deviation 0.55 on the log scale. For one job, it predicts $\widehat{\log(\text{cost})} = 6.2$.

What median cost would you quote the customer? What expected cost would you budget for 2,000 similar jobs? Calculate both.

Your turn

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What median cost would you quote the customer? What expected cost would you budget for 2,000 similar jobs? Calculate both.

- Median: $e^{6.2} = \$493$, a typical cost to quote.
- Mean: $e^{6.2+0.55^2/2} = \$493 \times 1.163 = \$573$. The expected budget is $2{,}000 \times \$573 = \$1.15\text{M}$. Using the median would give $\$0.99\text{M}$.

Revisiting the moving job from Unit 4

Recall the job with 800 cubic feet, a crew of 3, two flights of stairs, 12 miles, and packing.

	95% interval	Width
Confidence interval, average job like this one	12.25 to 12.54	0.3 hours
Prediction interval, this one job	10.74 to 14.04	3.3 hours

- For scheduling this job, we used the wider prediction interval because individual jobs vary around their average.
- We can now explain the 3.3-hour width and use the model to estimate probabilities within and beyond those endpoints.

Finding the prediction spread from the interval

A 95% prediction interval is $\hat{y} \pm t^* \, \text{SE}_{\text{pred}}$. Rearranging gives

$$\text{SE}_{\text{pred}} = \frac{14.04 - 10.74}{2 \times 1.9640} = \mathbf{0.839} \quad \text{hours.}$$

- The interval gives two percentiles of the forecast distribution. With this sample size, we can approximate it by

$$\text{hours for this job} \sim \text{Normal}(12.39, \, 0.839^2).$$

Key idea: using the full forecast

A regression model, together with its assumptions about residual variation, lets us describe a distribution of possible outcomes for the next case.

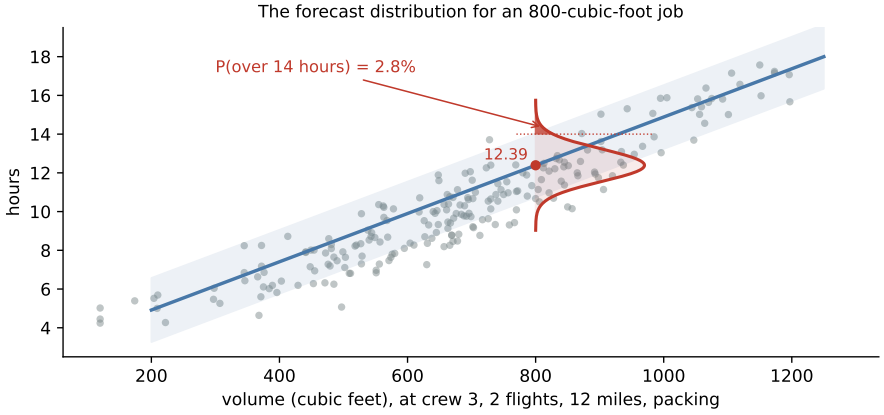
Why the prediction interval is wider

The model's residual standard deviation is **0.836**, close to the prediction spread of **0.839**. The prediction spread includes both uncertainty about the fitted mean and variation among individual jobs:

$$\mathrm{SE}^2_{\text{pred}} = \underbrace{\mathrm{SE}^2_{\text{line}}}_{0.073^2,\ \text{uncertainty about the mean}} + \underbrace{\sigma^2}_{0.836^2,\ \text{variation among jobs}}$$

- With 600 jobs, the uncertainty about the mean is small. Individual variation explains most of the prediction interval's width.
- **More data reduce uncertainty about the mean.** They do not remove the variation among jobs with the same predictors.

The forecast distribution for this job



The band shows prediction intervals across job volumes, holding the other predictors fixed. The sideways curve at 800 cubic feet shows the forecast distribution for this job. The shaded tail represents jobs lasting more than 14 hours.

Probabilities and a scheduling percentile

Question	z	**Answer**
$\mathbb{P}$(job runs past 13 hours)	0.73	23.4%
$\mathbb{P}$(job runs past 14 hours)	1.92	2.8%
$\mathbb{P}$(job runs past 15 hours)	3.11	0.09%
Time that 90% of similar jobs finish within	1.28	**13.47**

- The prediction interval summarizes a range. Using the full distribution, we can also estimate the chance of exceeding a particular time.
- The last row helps with scheduling: allowing 13.47 hours would cover about 90% of similar jobs under this model.

Planning for the chance of overtime

The crew starts earning overtime after a 13-hour shift.

- The **point prediction is 12.39 hours**, so the average job finishes before overtime begins.
- The forecast also gives a **23% chance of exceeding 13 hours**. Roughly one in four similar jobs would involve some overtime.
- Knowing that chance helps us plan staffing and costs, even though the point prediction is below the cutoff.

In practice: reporting the forecast

"We expect about 12.4 hours. There is about a 23% chance of going over 13 hours and a 3% chance of going over 14." This gives the dispatcher both an estimate and a sense of the risk.

A forecast from the logged diamond model

Unit 5's model was $\widehat{\log(\text{price})} = 7.90 + 1.88\log(\text{carat}) + \text{categories}$. For a 1.00 carat Ideal / G / VS2 stone:

Prediction on the log scale, $\hat{y}$	8.602
$\exp(\hat{y})$, the **median** price	\$5,443
The **mean** price, $\exp(\hat{y} + \sigma^2/2)$	\$5,492
95% prediction interval	\$4,179 to \$7,089

- The interval is symmetric on the log scale. After exponentiating, it is asymmetric around \$5,443.
- Calculate the endpoints on the log scale, then exponentiate them. Keep track of the scale of the prediction and its standard error.

The chance a diamond exceeds the budget

A buyer has a budget of $6,000 for this stone.

$$\mathbb{P}(\text{price} > 6000) = \texttt{lognorm.sf(6000, s=0.135, scale=5443)} = \mathbf{23.5}\%.$$

- We use the lognormal forecast because the model describes price on the log scale. A symmetric normal curve centered on $5,443 would give a different probability.
- As with the moving job, we identify the forecast distribution and calculate the area beyond the threshold we care about.

Key idea: the same approach for both models

The fitted prediction and residual variation give us a forecast distribution. Its tail areas help answer questions about overruns and budgets.

Total time for three jobs

Suppose we need the total time for three jobs like this one, $T = Y_1 + Y_2 + Y_3$. If the times are independent with the same normal forecast, Unit 7's rules give:

$\mathbb{E}[T] = 3 \times 12.39$	37.17 hours
$\mathrm{SD}(T) = 0.839 \times \sqrt{3}$	1.45 hours
$\mathbb{P}(T > 40)$	**2.6%**

- Add the *variances*, then take the square root. Adding standard deviations would give 2.52 hours and a probability of 13%, about five times as large.
- The independence assumption matters. Jobs sharing a crew, a truck, or the same weather may have related delays, which would change the total's spread.

Estimating overtime cost by simulation

Overtime costs \$85 an hour after 13 hours. What is the expected overtime cost for this job?

- Using only the prediction of 12.39 hours gives \$0. But some jobs run longer. This is the issue from Unit 7: for a nonlinear cost rule, $\mathbb{E}[g(Y)] \neq g(\mathbb{E}[Y])$ in general.
- Using the full distribution gives expected overtime of $\mathbb{E}[(Y-13)^+] = 0.115$ hours, or **\$9.77** per job. Across a thousand similar bookings, that is about \$9,800.

In practice: simulating the cost

```
sim = norm(12.39, 0.839).rvs(100000)
(85 * np.maximum(sim - 13, 0)).mean()
```

Simulate possible job times, calculate each overtime cost, and average the costs. We can use the same approach for more complicated pricing rules.

When the forecast needs more work

These calculations assume normal residuals with roughly constant spread. Problems with those assumptions can affect the probabilities we report.

- **A heavy right tail:** the model may underestimate long overruns. Check the upper end of the residual Q-Q plot.
- **Increasing spread:** if larger jobs vary more, one common residual SD can give intervals that are too narrow for large jobs and too wide for small ones.
- **Extrapolation:** a 2,000 cubic foot job is outside our data. Software can still produce an interval, but we have little evidence that the model works at that size.

Watch for: uncertainty about tail probabilities

A 1-in-20 overrun estimate depends on relatively few unusual outcomes. Give the tail a careful check before using that estimate to make a commitment.

Your turn

Your turn: four minutes, in pairs

A delivery-time model predicts 48 minutes for one route, with a 95% prediction interval of 34 to 62 minutes. The customer has been promised delivery within 60 minutes.

1. What is $\mathrm{SE}_{\mathrm{pred}}$? Use a normal approximation.
2. What fraction of similar deliveries would miss the promise under this model?
3. The residual plot suggests a heavier right tail than normal. How might that affect your estimate?

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$\mathrm{SE}_{\text{pred}} = (62 - 34)/3.92 = 7.14$. Then $z = (60 - 48)/7.14 = 1.68$, giving about **4.6%** late deliveries. A heavier right tail could make the actual late rate higher. We would check that tail before relying on the estimate.

A checklist for using a probability model

1. **Describe the outcome.** Are we counting successes, counting events over a period, measuring a wait, or adding smaller contributions? Use that context to suggest a model.
2. **Identify the assumptions.** Consider independence, a constant rate, or whether one contribution dominates. Which assumption needs the most careful check?
3. **Estimate the parameters.** Maximum likelihood gives familiar estimates in simple cases. Curvature near the log-likelihood peak can give approximate standard errors.
4. **Check the fit.** For Poisson counts, compare variance with mean and observed with expected counts. For continuous data, inspect the Q-Q plot.
5. **Answer the practical question.** Calculate the probability, percentile, or expected cost that would help someone make a decision.
6. **Explain the limits.** Say which part of the distribution your answer depends on and how well the model fits there.

Distributions at a glance

Distribution	**Typical setting**	**Mean**	**Useful check**
Binomial(n, p)	Successes in n fixed tries	np	Observed vs. expected counts
Poisson(λ)	Events per unit, no fixed limit	λ	Is variance close to mean?
Exponential(λ)	Wait until the next event	$1/\lambda$	Is the event rate constant?
Normal(μ, σ^2)	Many small contributions added	μ	Q-Q plot, including relevant tails
Lognormal	Multiplicative effects	$e^{\mu+\sigma^2/2}$	Q-Q plot of $\log y$

Software reference

Task	How
Fit a named distribution	`scipy.stats.<dist>.fit(x)`
Expected counts under a fit	`poisson.pmf(k, lam) * n`
Check Poisson dispersion	`x.var() / x.mean()`
Make a Q-Q plot	`scipy.stats.probplot(x, plot=ax)`
Find a probability or percentile	`.cdf()`, `.sf()`, `.ppf()`
Recover $\mathrm{SE}_{\text{pred}}$ from an interval	Width divided by $2t^*$
Approximate a normal forecast	`norm(yhat, se_pred)`
Find a lognormal model's mean	`np.exp(yhat + sigma**2/2)`

Common mistakes to watch for

- Choosing a distribution from a histogram alone.
- Assuming that a successful software fit means the model fits well.
- Using a Poisson model despite much more variation than it allows.
- Treating a small goodness-of-fit p-value as the only measure of usefulness.
- Reading a density height as a probability.
- Reporting $\exp(\hat{y})$ from a lognormal model as the mean.
- Making symmetric intervals on the original scale for a logged model.
- Reporting a tail probability after checking only the middle of the distribution.

Questions for review

1. How does a probability model go beyond a description of the observed data?
2. When would you consider a binomial, Poisson, exponential, or normal distribution?
3. What does a Poisson model assume about variance, and how can you check it?
4. What is likelihood? What does a maximum likelihood estimate maximize?
5. How does the shape of the log-likelihood relate to a standard error?
6. What are the terms in AIC? Why do models for y and $\log y$ need an adjustment before comparing their reported AIC values?
7. What is overdispersion, and how can it affect standard errors?
8. Why is $\mathbb{P}(X = 10.43) = 0$ for a continuous outcome?
9. A lognormal model predicts $\hat{y}$ on the log scale. How do you get the median and mean?
10. How can a prediction interval help you estimate the chance of exceeding a threshold?

Where we go next

- We began by fitting a distribution to one variable at a time: one weekend probability for all 600 jobs, or one stairs rate for every job.
- Unit 9 adds predictors for binary outcomes and counts. Logistic regression models a yes-or-no probability. Poisson regression lets an expected count vary with predictors.
- Both use likelihood to estimate their parameters, so the ideas from this unit carry directly into those models.
- We will also return to overdispersion and check whether the count model gives realistic standard errors.